Classwork 3: Generating data from a nested logit model

Figure: Morrells choice beers, Malt Stout, Castle Ale and College Ale

Classwork 3 I

- $T = 1000$ markets, $J = 10$ products, $R = 2$ nests + outside good
- Observed characteristics and unobserved utility:

$$\xi_{jt} \sim 0.5 * \mathrm{N}(0,1)$$
$$x_{jt} \sim 0.5 * \mathrm{Uniform}[0,1]$$

- Nested-logit (GEV) preferences with mean utility

$$\delta_{jt} = -2 + x_{jt} - p_{jt} + \xi_{jt} \qquad \delta_{0t} = 0$$

- Nesting parameter $\sigma = 0.5$
- Marginal cost:

$$c_{jt} = x_{jt} + 0.5 * \omega_{jt} \qquad \omega_{jt} \sim 0.5 * \mathrm{Uniform}[0,1] \tag{1}$$

- j allocated to group $r = (1,2)$ randomly (allocation iid across markets).

Classwork 3 II

- Conduct: (single-product-firm) Nash pricing.
- Generate endogenous prices and market shares with Nested logit market shares and single-product Nash pricing using a nonlinear equation solver.
- Invert market share system analytically and estimate by OLS.
- Estimate by IV instrumenting only for price (using marginal cost); show this is not sufficient.
- Generate the instrument for the within-group market share: a count of the number of goods in each market-group.
- Estimate by IV.
- Calculate an own- and cross-price elasticity matrix to assess additional flexibility relative to simple logit.

See suggested code on Canvas.